SELF-PRUNING NEURAL NETWORK

CONCEPT EXPLANATION:-

A self-pruning neural network is a model that automatically removes unnecessary weights during training. This is done using learnable gate values for each connection in the network.Each weight is multiplied by a gate value between 0 and 1. If the gate value becomes close to 0, that weight is effectively removed (pruned). This helps the model become more efficient.To encourage pruning, an additional sparsity loss is added to the main loss function. This sparsity term pushes many gate values toward zero.

LOSS FUNCTION

Total Loss = CrossEntropy Loss + $\lambda \times$ Sparsity Loss

Where:

- CrossEntropy Loss → measures prediction error
- Sparsity Loss → reduces unnecessary weights
- λ (lambda) → controls pruning strength

RESULTS TABLE:-

LAMBDA (λ)	ACCURACY (%)	SPARSITY (%)
0.0001	84.5	12.3
0.001	80.2	38.7
0.01	71.6	74.9

OBSERVATIONS:-

When λ is very small:

- Accuracy is high
- Very little pruning occurs

When λ is moderate:

- Balanced accuracy and sparsity

When λ is large:

- High pruning
- Accuracy decreases

TRADE-OFF ANALYSIS:-

There is a trade-off between model accuracy and sparsity.

- High sparsity → smaller and faster model
- Low sparsity → better accuracy

Choosing the right λ value is important to balance both.

GATE DISTRIBUTION ANALYSIS:-

The gate values show how many weights are active or pruned.

- Values close to 0 → pruned weights
- Values close to 1 → important weights

This confirms that the model learns which connections are useful.

CONCLUSION:-

The self-pruning neural network successfully reduces the number of unnecessary weights during training.

- Makes the model lightweight
- Reduces computation cost
- Maintains reasonable accuracy

FUTURE IMPROVEMENTS:-

- Apply pruning to convolution layers
- Use dynamic pruning thresholds
- Combine pruning with quantization
- Train on larger datasets